

Medical PPE Detection using YOLOv8



Project Motivation

- **Healthcare environments require strict Personal Protective Equipment compliance**
- **Manual monitoring is time-consuming**
- **Computer vision can automate PPE detection in real time**
- **Goal: Build an accurate and efficient detection system**

01 Dataset Acquisition & Exploration

- Dataset contains annotated medical PPE images
- Multiple object classes: masks, gloves, gowns, caps, shoe covers
- Exploratory analysis performed on class distribution and image quality
- Scraping Data from different websites

caps



masks



gloves



shoe
covers



gowns

03 Dataset Cleaning & Revision

Revising the Collected Data

- After scraping, the dataset contained unnecessary and low-quality images
- Performed manual and automated filtering to improve dataset quality

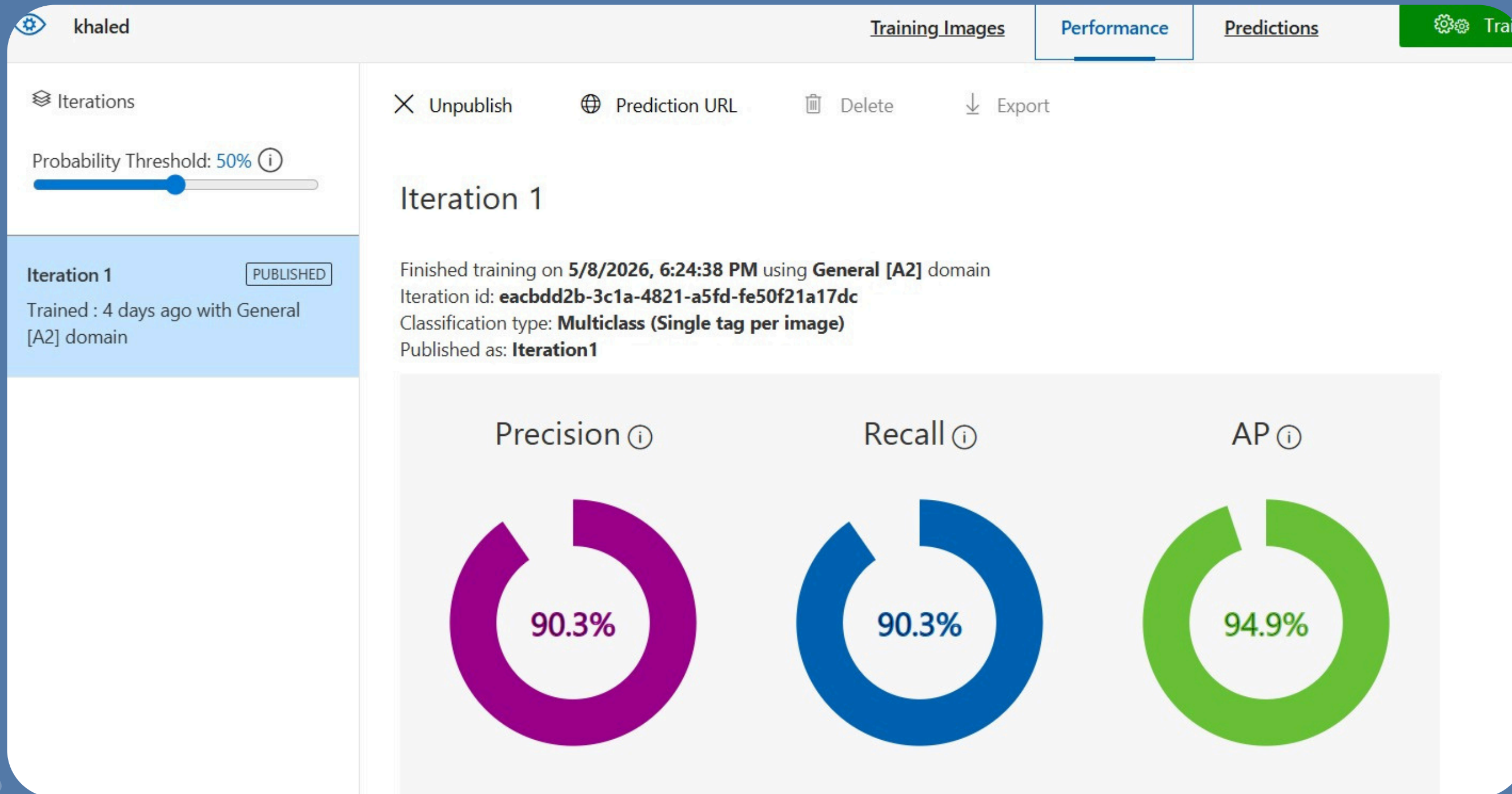
Removed Images That:

- Did not contain PPE objects
- Had blurry or low-resolution quality
- Included incorrect or irrelevant content
- Contained duplicate samples
- Had inaccurate annotations

Result

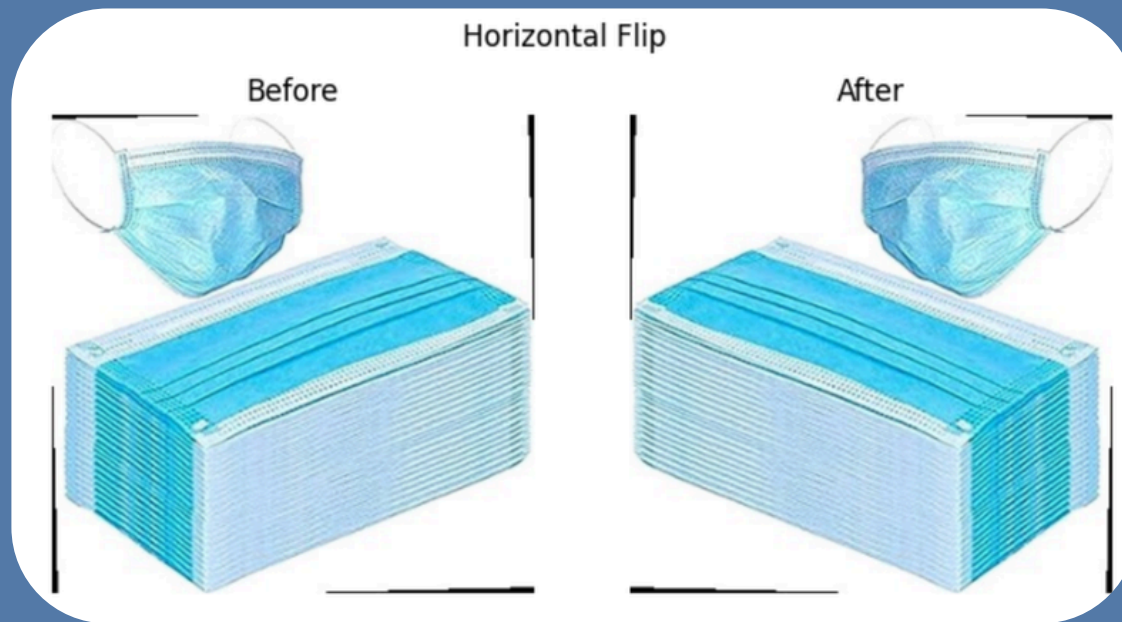
- Cleaner and more reliable dataset
- Reduced noise during training
- Improved model accuracy and generalization

03 Dataset Cleaning & Revision

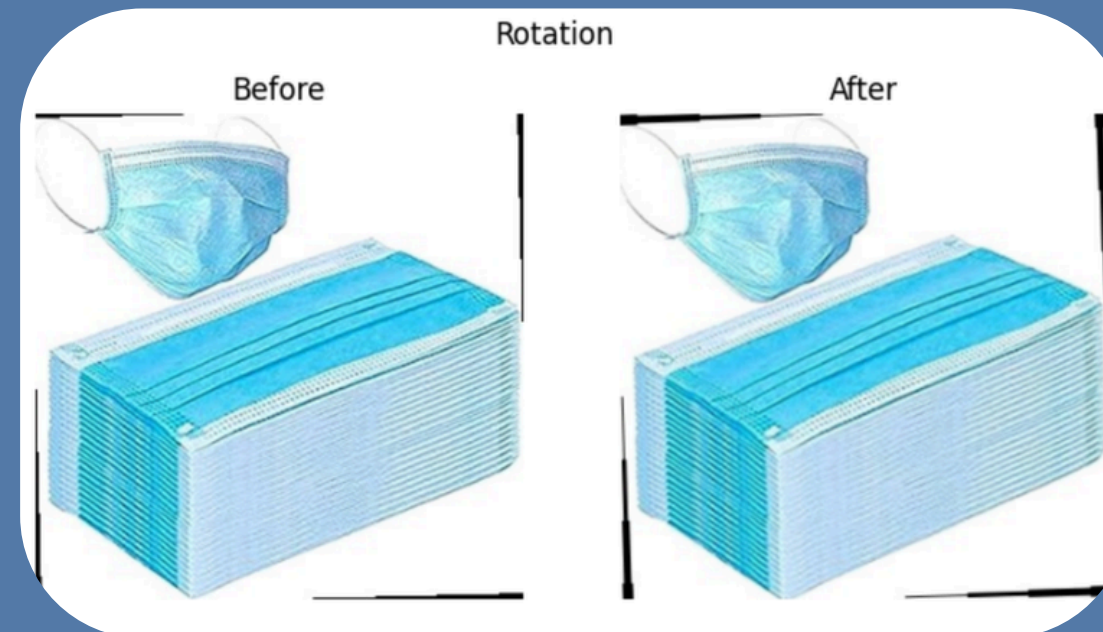


04 Data Augmentation Techniques

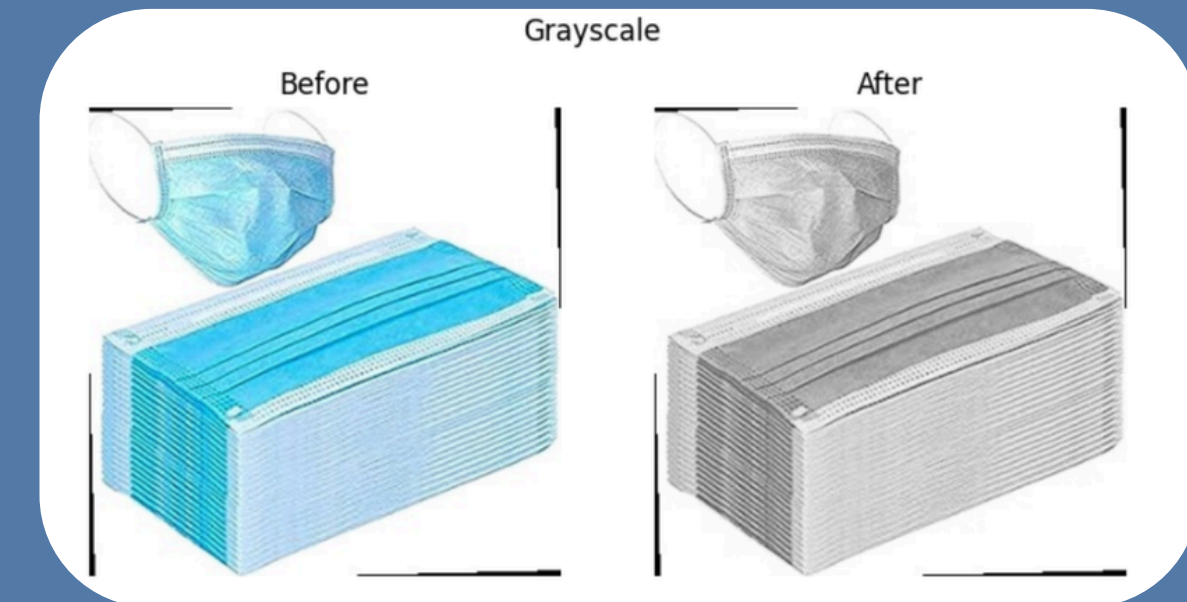
1. Horizontal flipping



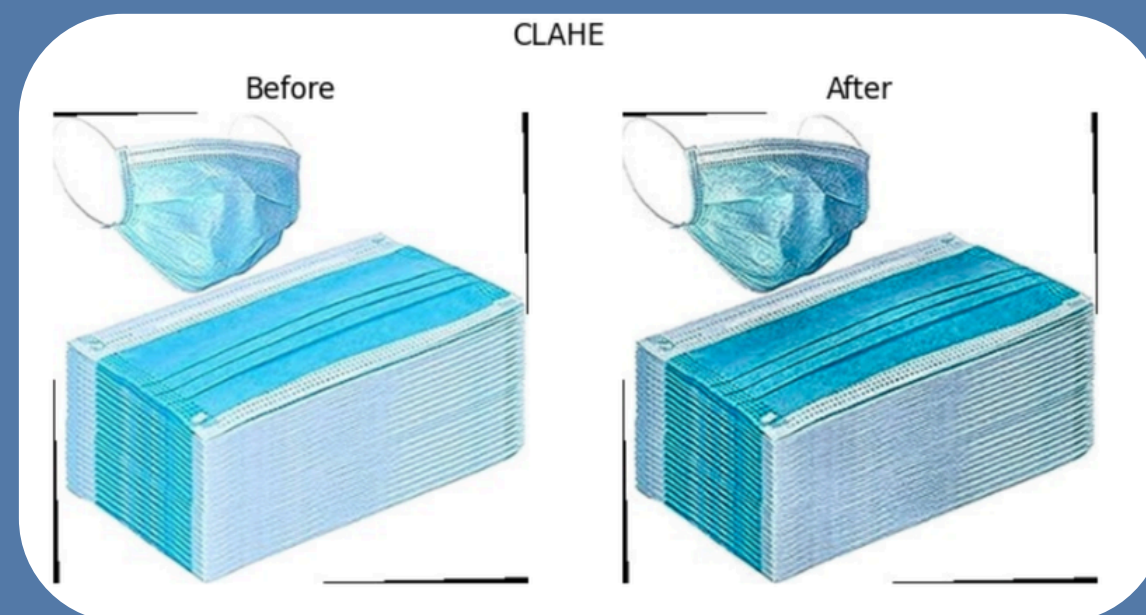
2. Rotation



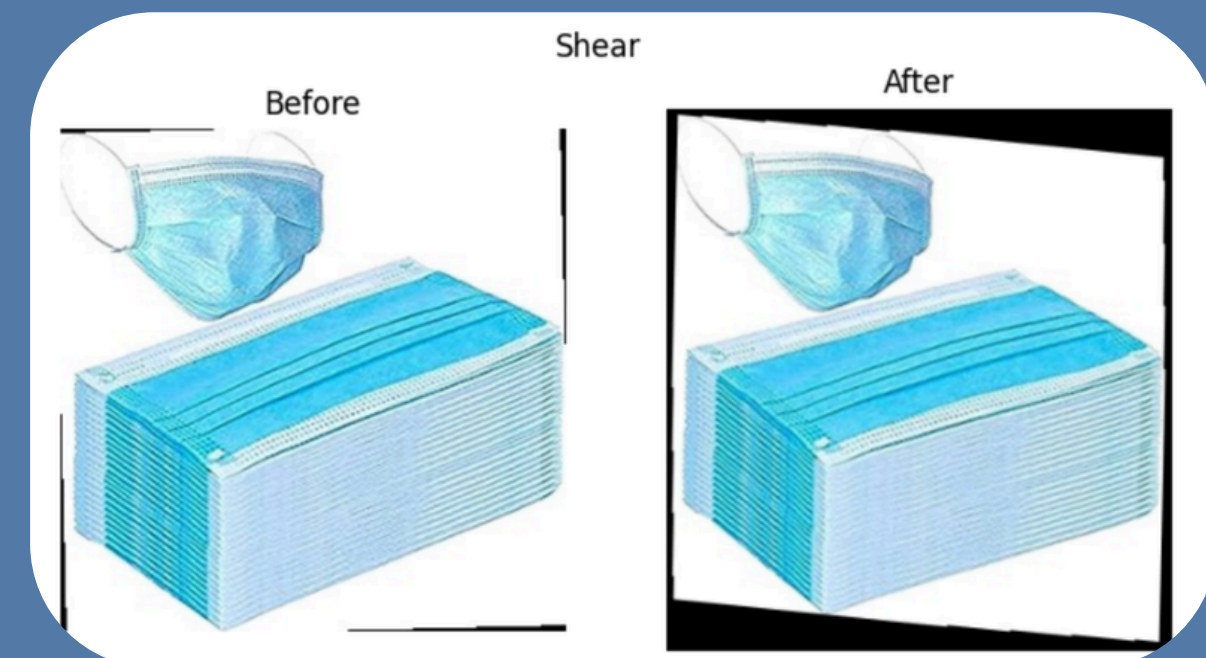
3. GrayScale



4. CLAHE



5. Shear



05 YOLOv8 Training Pipeline

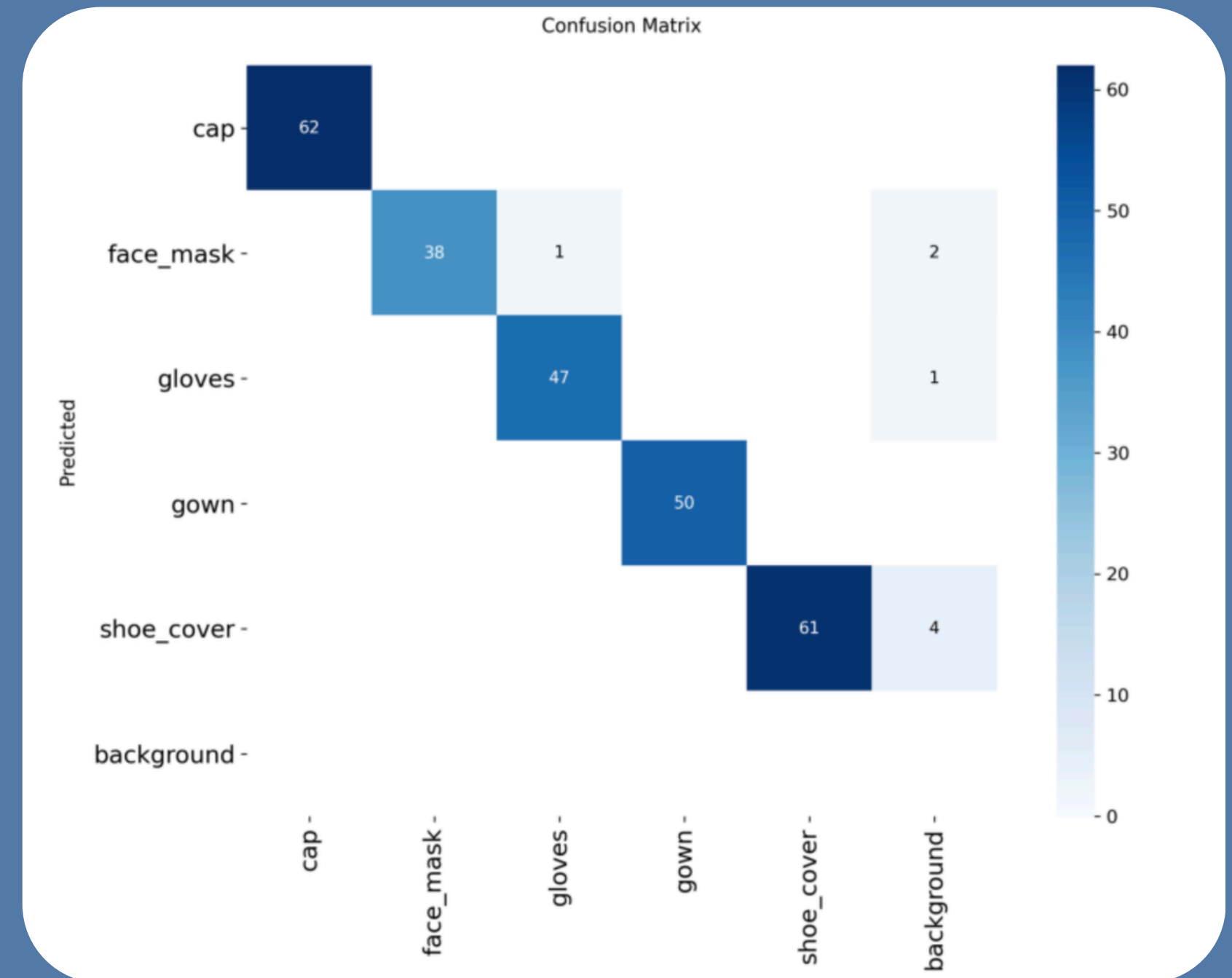
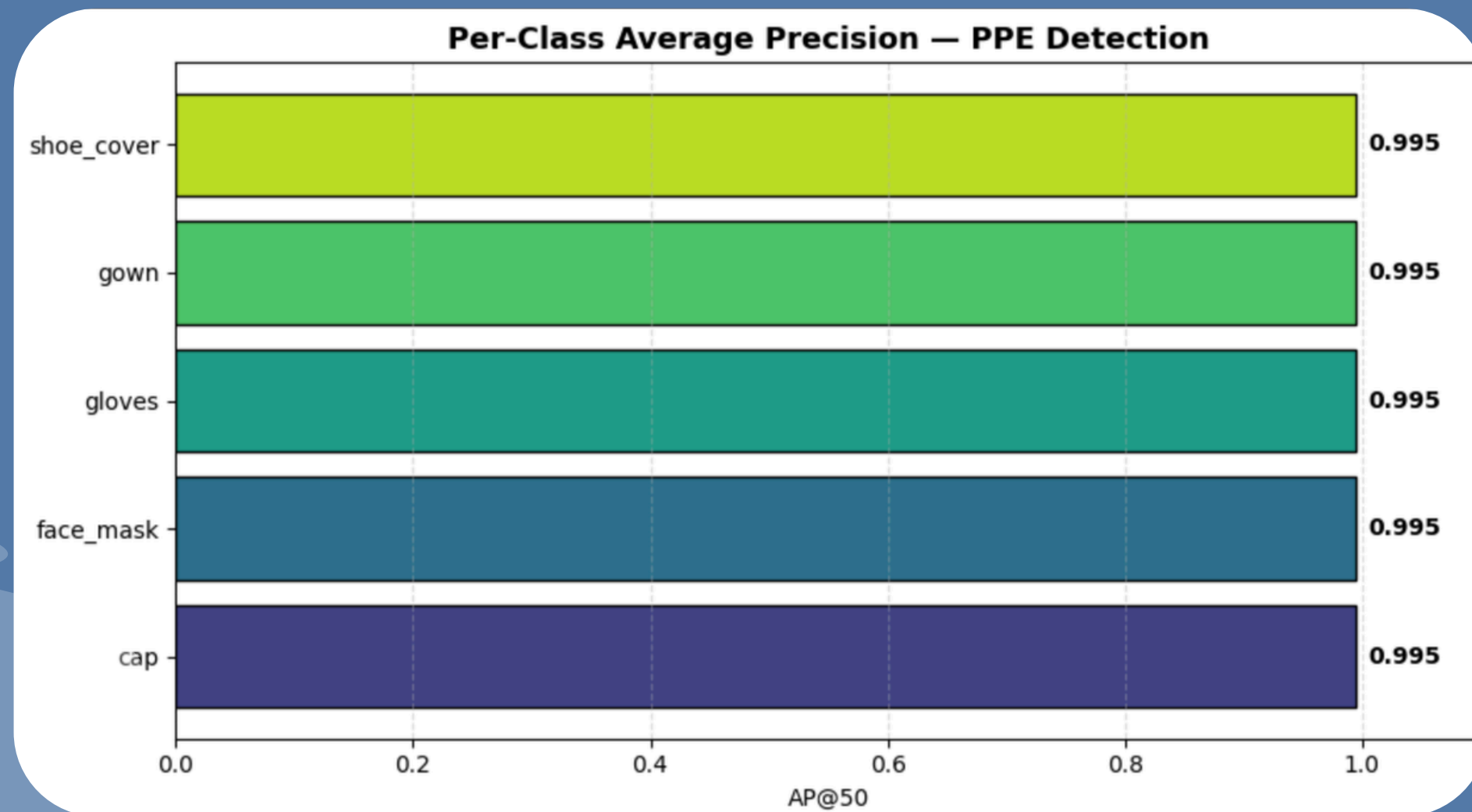
- Trained two configurations for comparison
- Configuration 1: YOLOv8n Baseline
- Configuration 2: YOLOv8s Tuned Model
- Compared learning rate, batch size, and augmentation settings
- WINNER: Config2_YOLOv8s_Tuned
- $\text{mAP@50} = 0.995$



06 Hyperparameter Tuning

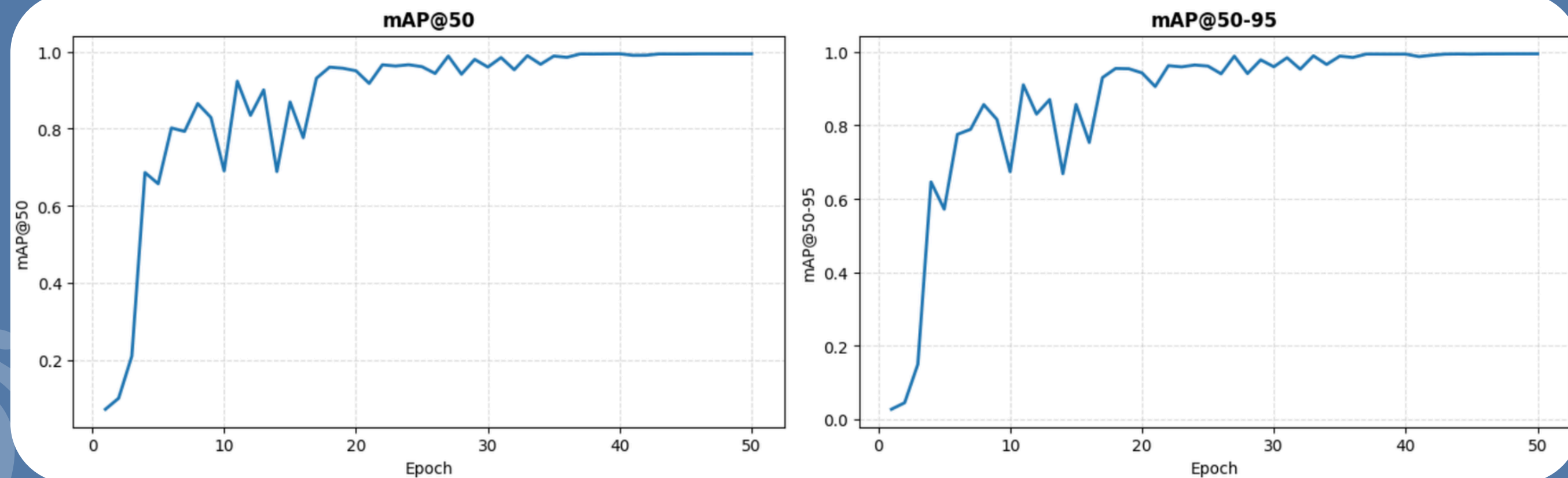
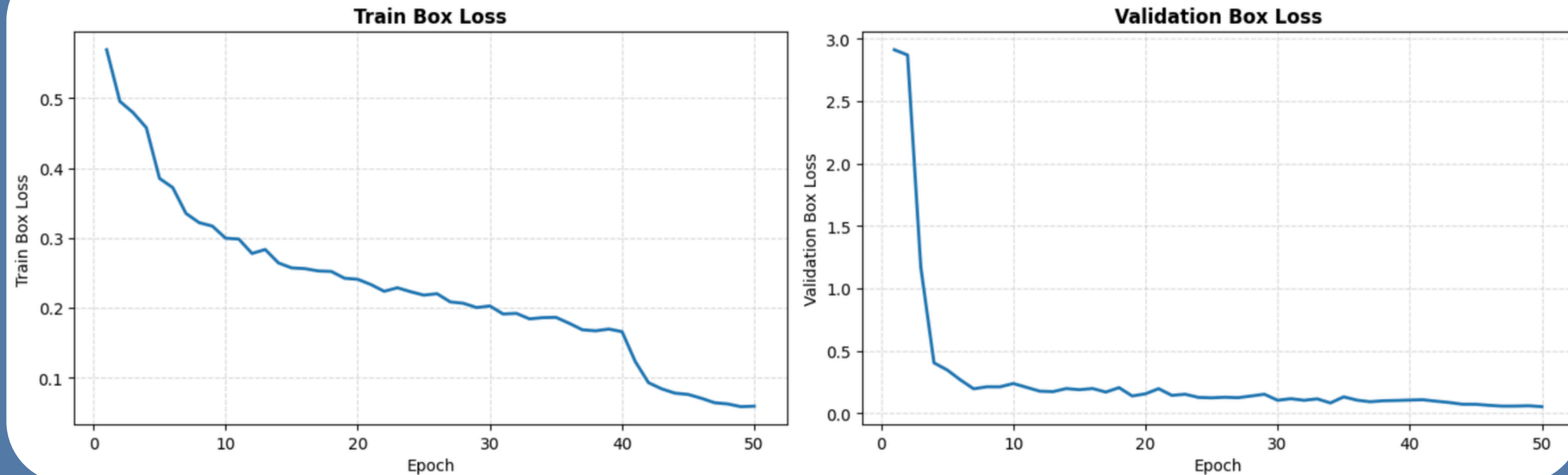
- Adjusted learning rate for stable convergence
- Modified batch size for hardware efficiency
- Applied augmentation probabilities during training
- Selected best model based on accuracy and inference speed

07 Model Evaluation

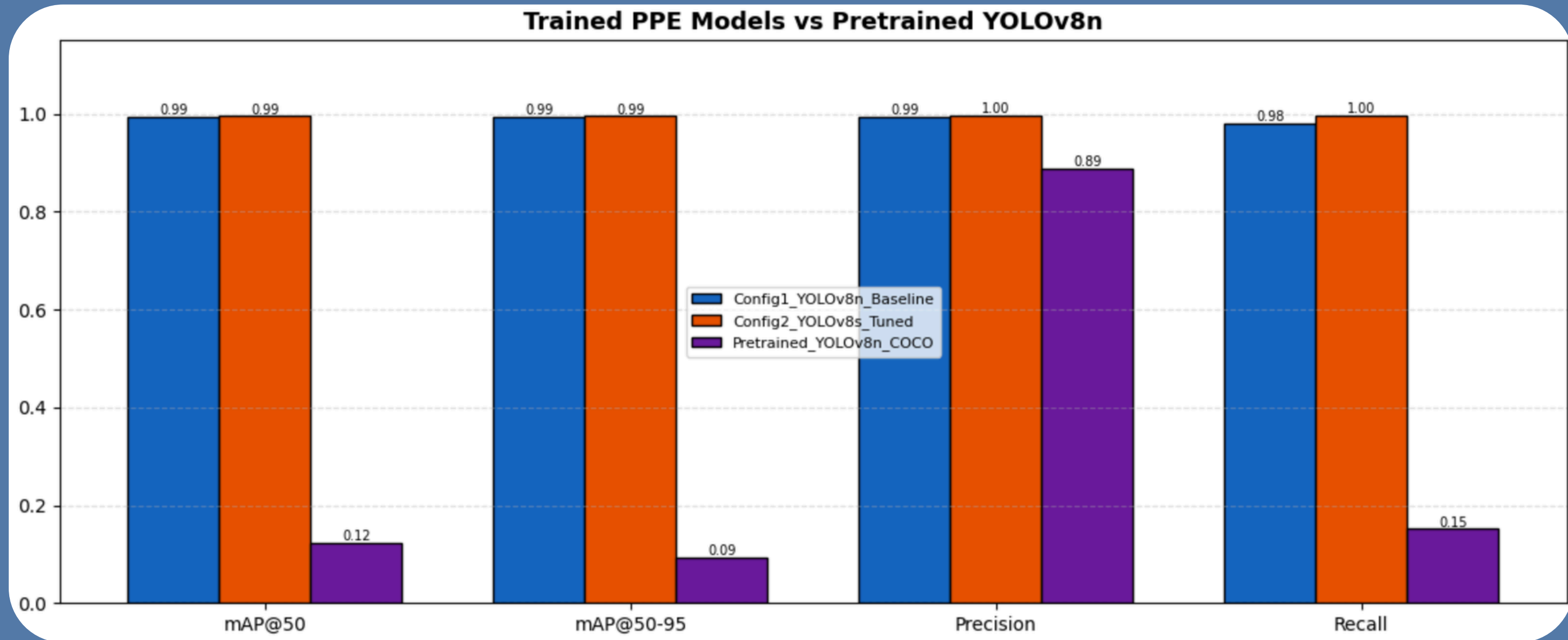


08 Training Curves

Training Curves — Config2_YOLOv8s_Tuned



09 Compare with Pretrained YOLOv8

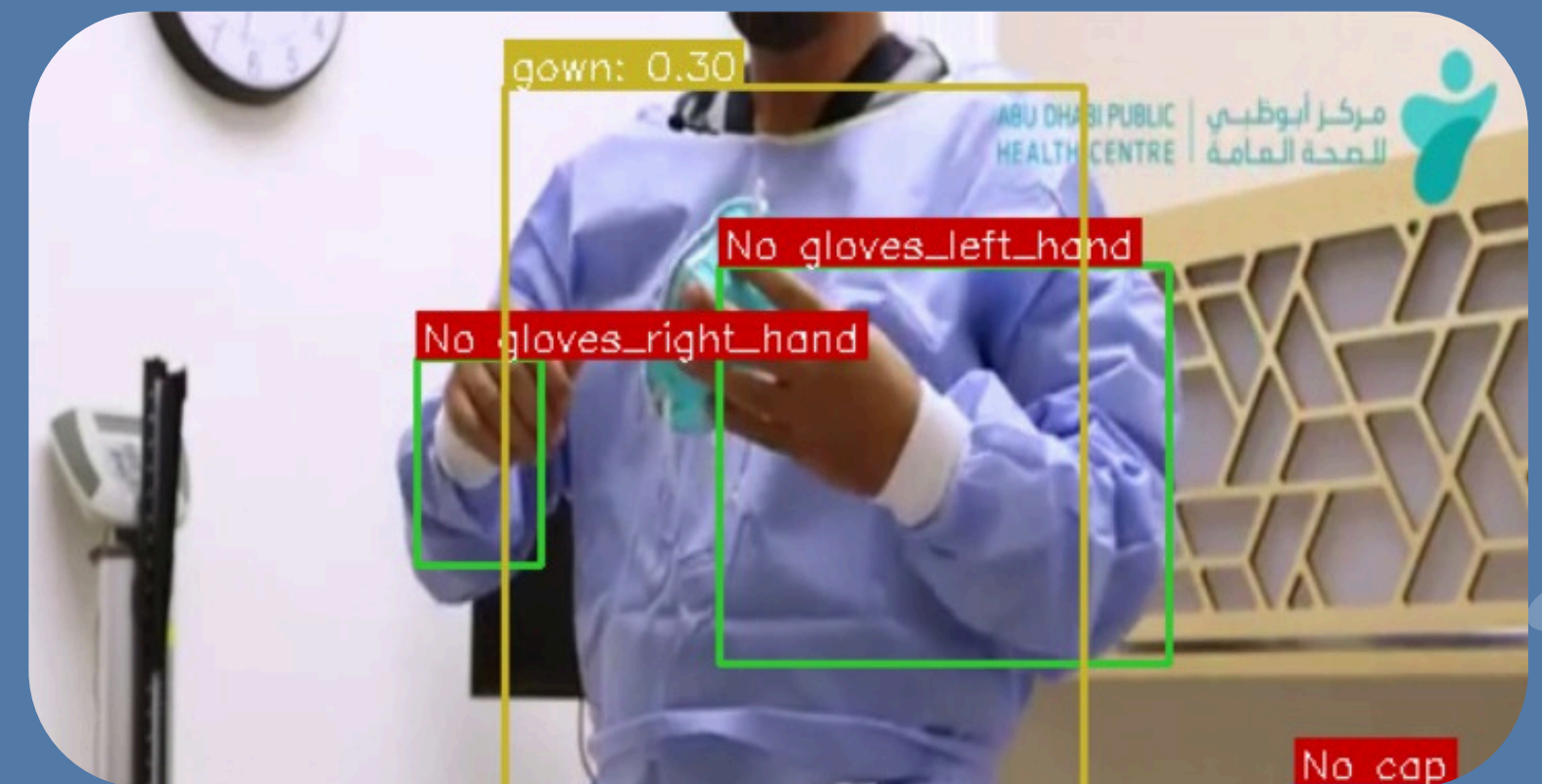
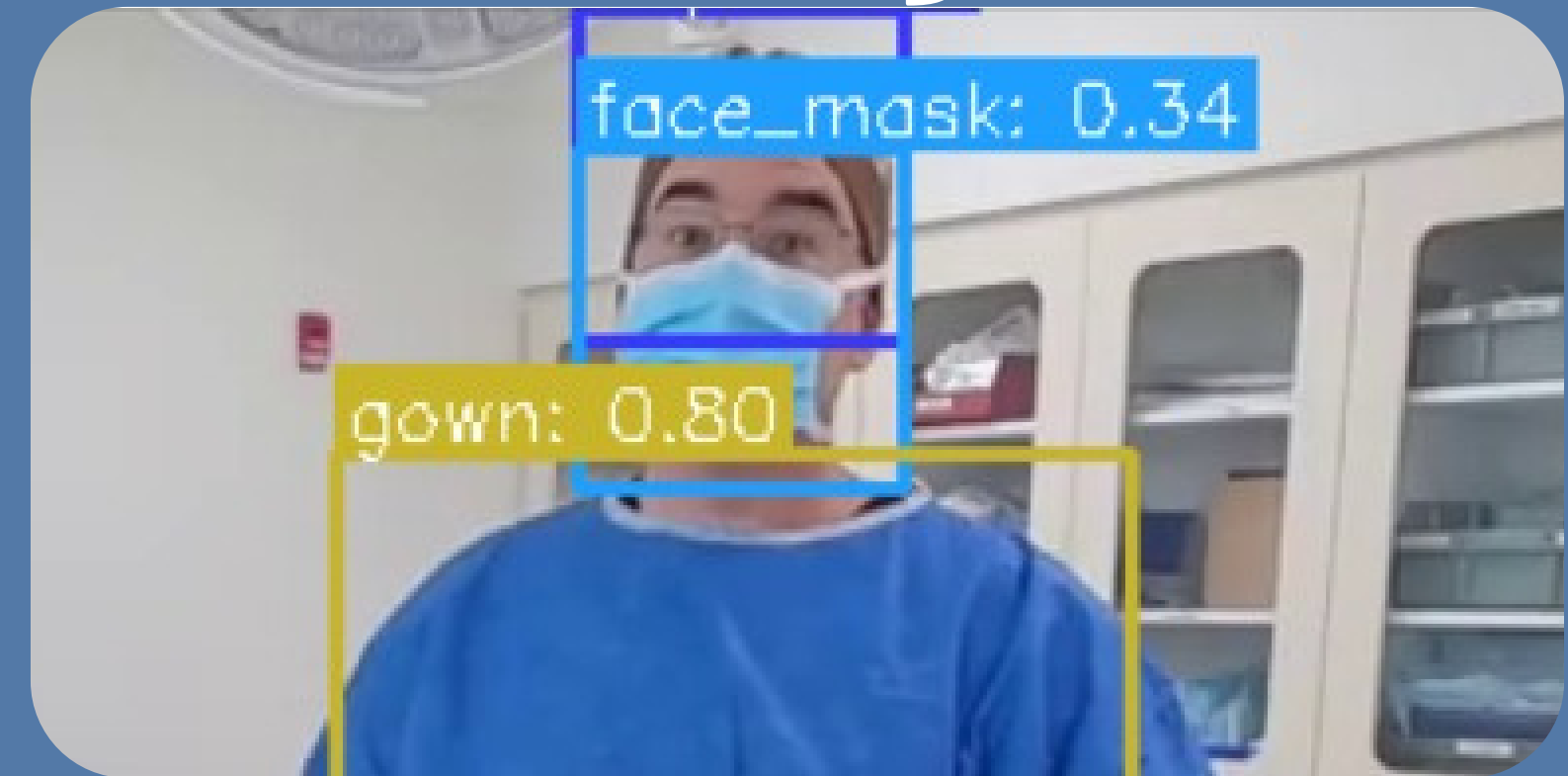


10 ROI-Based Detection Improvement

- Solving the Full-Image Detection Problem
- One challenge with YOLO is that it processes the entire image at once
- Small PPE objects may be missed because they occupy a limited area in the frame
- To improve detection accuracy, the image was divided into multiple Regions of Interest (ROIs)
- Each ROI was analyzed separately to search for the PPE item expected in that specific region
- Example:
 - Upper region → face mask and cap detection
 - Middle region → gown detection
 - Lower region → gloves and shoe cover detection
- This approach improved focus on smaller objects and reduced false detections
- ROI-based processing increased overall detection reliability in real-time scenarios

11 Real-Time Detection System

- Implemented using OpenCV and YOLOv8
- Supports webcam and video input
- Displays FPS and confidence threshold
- Can trigger alerts for missing PPE



Conclusion

- **Developed a complete YOLOv8-based visual recognition pipeline**
- **Achieved real-time PPE detection with strong performance**
- **Demonstrated practical AI application in healthcare safety**
- **Project combines preprocessing, training, evaluation, and deployment**



THANK YOU

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